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To the Editor-in-Chief and Associate Editor
IEEE Transactions on Information Theory

Dear Editor:

Please consider the enclosed manuscript, "Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context," for publication in the IEEE Transactions on Information Theory. I am the sole author.

The paper evaluates Steinberg's common-reconstruction functional on an augmented Gaussian source: the encoder observes a pair (Y, V), the distortion binds Y alone, and a third party holds a noisy copy of the context V that no decoder uses. The core results are a closed form for the minimal conditional content in any ambient dimension, the larger root of an explicit quadratic; the exact region of jointly achievable rate and conditional content, with a two-water-level Pareto frontier; and a complete elementary solution of the binary case through a tilt equation. Structural consequences: a strict rate–content tradeoff opens exactly when the described variable and the context are partially correlated, and persists at the clean-context boundary; the determinant lower bound is attained exactly on two branches governed by the encoder-accessible conditional covariance; and the function is not determined by the reduced source–side-information marginal. Extensions give the frontier with vector context and a direct Gaussian operational theorem. Consistency anchors recover the classical, Gray, and Steinberg corners and identify the weighted objective as an information-bottleneck Lagrangian augmented by a hard distortion constraint.

Two earlier documents are relevant, and they are distinct from one another. First, this Transactions recently declined a different manuscript of mine, a broad synthesis. I ask the editors to read the present submission independently of it: this is a single-object work in the classical rate–distortion genre and shares no theorem with that declined submission. The terminology concerns raised in that review have been addressed here by adopting standard source-coding vocabulary throughout.

Second, and separately, the manuscript cites an archived preliminary manuscript by the author (Zenodo record, doi:10.5281/zenodo.21776291), as disclosed in its Related Work section. The discrete operational region theorem and its single-letterization appeared there, as did the scalar Gaussian corner, and both are inherited here with attribution. Everything from Section III onward is new, and the present paper supersedes the source-coding content of that record, which remains the archived evidence artifact.

All closed forms are verified by an apparatus available as supplementary repository material: symbolic and numerical verification scripts, including two written independently of the derivations, and a partial Lean 4 formalization of the load-bearing algebra.

This manuscript is not under review elsewhere, and I have no conflicts of interest to declare.

Sincerely,

Andrew H. Bond
Senior Member, IEEE